Monthly Research Trends: Developmental Cognitive Neuroscience

Report Date: 2026-05-17
Period Covered: 2026-05 (May 2026; recent 2026 publications included)
Total Papers Found: 18

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1. Executive Summary

The following five findings represent the most significant contributions to the field this month and in recent weeks:

1. **Neonatal fMRI predicts 18-month socioemotional outcomes** *(bioRxiv, May 2026)*: Using connectome-based predictive modeling on 397 neonates from the Developing Human Connectome Project, this preprint demonstrates that whole-brain resting-state connectivity at birth can prospectively predict externalizing behavior, surgency, and effortful control at 18 months. This matters because it moves infant neuroimaging from descriptive to prognostic, with direct implications for early intervention targeting.

2. **Explicitly nonlinear fMRI reveals hidden infant developmental trajectories** *(bioRxiv, April 2026)*: The first whole-brain nonlinear fMRI connectivity analysis in typically developing infants uncovered networks associated with default mode processes, executive function, and language production that were invisible to conventional linear methods. This methodological advance reframes how we conceptualize and measure early functional organization.

3. **Single-cell atlas of the developing Down syndrome cortex** *(Nature Medicine, January 2026)*: Profiling ~250,000 cells from 15 DS and 15 control fetal cortices identified subtype-specific reductions in excitatory neurons and pinpointed three chromosome 21 transcription factors

(BACH1, PKNOX1, GABPA) as dosage-sensitive hubs. This is the highest-resolution molecular map of the DS brain to date and opens therapeutic targets for antisense oligonucleotide intervention.

4. **Aperiodic EEG exponent links early E/I balance to restricted and repetitive behaviors in autism** *(J. Neurodevelopmental Disorders, 2025/2026)*: Lower aperiodic exponent at 12–14 months specifically predicted greater restricted and repetitive behaviors (RRBs) through toddlerhood, independent of general cognitive development. This positions the aperiodic exponent as a domain-specific biomarker for ASD behavioral trajectories.

5. **Cascading sensitive periods for language-related brain plasticity** *(bioRxiv, March 2026)*: Using Hurst exponent as an in vivo index of cortical plasticity across the Baby Connectome Project (10 months–3.5 years) and HCP-Development (5–15 years), the study maps a posterior-to-anterior cascade of language-sensitive periods and shows that stronger language skills are paradoxically associated with slower cortical maturation—reflecting extended plasticity as an advantage.

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2. Trending Topics This Month

Most active areas (May 2026 publication landscape):

- **EEG aperiodic activity** is the single most prolific subdomain: multiple independent groups are applying 1/f aperiodic decomposition (FOOOF/specparam) to infant and toddler datasets, linking the exponent and offset to autism risk, language trajectories, nutrition, and maternal mental health. This represents a methodological shift away from band-power approaches.
- **Resting-state fMRI in neonates and infants** is surging, with studies using dHCP data, nonlinear connectivity frameworks, and connectome-based predictive modeling—all converging on the finding that neonatal functional architecture is already predictive of later behavioral and cognitive outcomes.

- **Language network development** is receiving multi-method attention, with simultaneous EEG-fMRI in neonates, toddler fMRI, and Hurst-exponent plasticity studies all mapping when and where language-relevant neural organization emerges.
- **Sensory processing and E/I balance in autism** continues strong momentum, with longitudinal EEG cohorts enriched for familial autism/ADHD risk revealing divergent trajectories of sensory hyper- vs. hypo-responsivity.
- **Methodological advances**: awake infant fMRI methodology, nonlinear connectivity analysis, and deep learning for EEG/fNIRS-based ASD classification are all actively advancing.

Emerging themes: - Integration of environmental/nutritional exposures with neural biomarkers (e.g., breastfeeding and aperiodic slope). - Maternal mental health as a driver of infant EEG trajectory differences. - Connectome-based predictive modeling extending from structural to functional neonatal data.

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3. Papers by Topic

3.1 EEG in Infants & Toddlers

Paper 1
Title: Tracing infant sleep neurophysiology longitudinally from 3 to 6 months: EEG insights into brain development
Authors: Beaugrand M., Jaramillo V., Mühlematter C., et al.
Journal: *npj Biological Timing and Sleep*
Date: March 2026
Preprint? No (peer-reviewed)
Age Range: 3–6 months
Primary Method: EEG (sleep NREM)
Statistical Approach: Longitudinal EEG power analysis; scalp topographic mapping across NREM sleep; mixed-effects modeling
Key Findings: Slow-wave activity (SWA, 0.75–4.25 Hz) increased maximally in

occipital regions from 3 to 6 months, while theta power showed a global increase across all scalp regions. Sigma power, initially concentrated centrally, dispersed toward frontal regions over the same period—a pattern consistent with progressive thalamocortical maturation. These topographic developmental shifts were linked to behavioral developmental milestones.
DOI/URL: https://www.nature.com/articles/s44323-026-00071-7

Paper 2
Title: Effects of Early Infant Nutrition on Aperiodic Exponent From 2 to 12 Months Using EEG Analysis
Authors: Gilbreath M., et al.
Journal: *Psychophysiology*
Date: February 2026
Preprint? No (peer-reviewed)
Age Range: 2–12 months
Primary Method: High-density EEG (resting-state; silent video baseline)
Statistical Approach: Specparam/FOOOF aperiodic decomposition; source-space reconstruction; linear mixed models comparing breast milk vs. formula feeding
Key Findings: The aperiodic exponent decreased with age across all infants, consistent with established E/I maturation trajectories. Breastfed infants showed a significantly larger aperiodic exponent than formula-fed infants at 3 months only; no differences were found at 2, 4, 5, 6, 9, or 12 months. This is one of the largest infant EEG datasets to date and one of the first to apply source-space reconstruction for spatial mapping of aperiodic activity.
DOI/URL: https://onlinelibrary.wiley.com/doi/10.1111/psyp.70247

Paper 3
Title: EEG trajectories across the first two years of life are associated with maternal depression, anxiety, and perceived stress
Authors: [First author not confirmed from search; large community-based cohort]
Journal: *Journal of Affective Disorders* (ScienceDirect)
Date: 2026

Preprint? No (peer-reviewed)
Age Range: 2–24 months
Primary Method: EEG (longitudinal resting-state; high-frequency oscillations β and γ)
Statistical Approach: Longitudinal linear mixed models; maternal psychological distress scales (depression, trait anxiety, perceived stress)
Key Findings: Higher maternal depression symptoms, trait anxiety, and perceived stress were each associated with significantly lower beta and gamma power trajectories in infants from 2 to 24 months. This is reported as the first longitudinal study tracking EEG trajectories across the full first two years in relation to maternal mental health; the sample (N=211) included high proportions of Black/African American families (44.5%), increasing generalizability.
DOI/URL: https://www.sciencedirect.com/science/article/abs/pii/S0165032725020294

Paper 4
Title: Resting-state periodic and aperiodic brain oscillations from birth to preschool years: Aperiodic maturity predicts developmental course
Authors: [Multiple; longitudinal cohort study]
Journal: *Developmental Cognitive Neuroscience* (ScienceDirect)
Date: 2026
Preprint? No (peer-reviewed; earlier version on bioRxiv May 2025)
Age Range: 2–68 months
Primary Method: EEG (resting-state, source-localized; 7 brain regions)
Statistical Approach: Specparam aperiodic decomposition; dominant frequency analysis; mixed-effects nonlinear growth curve modeling; adaptive behavior regression
Key Findings: Aperiodic exponent and offset decreased nonlinearly with age and differed across brain regions; more mature (steeper) aperiodic values at earlier ages predicted better adaptive behaviors and daily living skills. Dominant periodic oscillations emerged more often during dark-room conditions (eliciting 36% more activity than video-on). This comprehensive birth-to-preschool EEG dataset establishes normative trajectories against which clinical populations can be benchmarked.

DOI/URL: https://www.sciencedirect.com/science/article/pii/S1878929326000447

3.2 fMRI in Infants & Toddlers

Paper 5
Title: Neonatal Resting-State Functional Connectivity Predicts Socioemotional and Behavioral Outcomes at 18 Months
Authors: [Developing Human Connectome Project consortium; preprint—full author list in bioRxiv]
Journal/Venue: *bioRxiv* (preprint)
Date: May 05, 2026
Preprint? Yes (bioRxiv)
Age Range: Neonates (term-born and preterm); outcomes at 18 months
Primary Method: Resting-state fMRI
Statistical Approach: Stability-driven, ROI-constrained connectome-based predictive modeling (CPM); whole-brain connectivity features; Child Behavior Checklist (CBCL) and Early Childhood Behavior Questionnaire (ECBQ)
Key Findings: Neonatal whole-brain functional connectivity significantly predicted 18-month externalizing behavior, surgency, negative affect, and effortful control in 397 infants (277 term, 120 preterm). Predictive architectures differed substantially between term and preterm groups, suggesting that prematurity fundamentally alters the neural substrates of socioemotional regulation. This is among the first studies to demonstrate neonatal fMRI as a prognostic tool for behavioral temperament outcomes.
DOI/URL: https://www.biorxiv.org/content/10.64898/2026.04.21.719787v2

Paper 6
Title: Explicitly nonlinear fMRI networks reveal hidden trajectories of infant brain development
Authors: [preprint, author list in bioRxiv]
Journal/Venue: *bioRxiv* (preprint)
Date: April 08, 2026

Preprint? Yes (bioRxiv)
Age Range: Typically developing infants (postnatal months, range not confirmed)
Primary Method: Resting-state fMRI (nonlinear connectivity)
Statistical Approach: Data-driven nonlinear functional connectivity decomposition; comparison with linear ICA-derived networks; developmental regression modeling
Key Findings: Linear and nonlinear fMRI network counterparts are linked to partially overlapping but complementary developmental profiles; the nonlinear approach uniquely revealed networks associated with default mode processes, executive functioning, language production, and stimulus saliency that were absent or obscured in standard linear analyses. This is the first comprehensive whole-brain nonlinear resting-state fMRI study in human infants.
DOI/URL: https://www.biorxiv.org/content/10.64898/2026.04.07.716703v1

Paper 7
Title: Functional Magnetic Resonance Imaging in Awake Infants: Insights from More Than 750 Scanning Sessions
Authors: Behm et al. (multi-lab: Yale and MIT)
Journal: *Infancy*
Date: 2026 (journal publication; bioRxiv preprint February 2025)
Preprint? No (peer-reviewed)
Age Range: 1–36 months
Primary Method: Awake task-based fMRI
Statistical Approach: Multi-lab retrospective analysis; logistic and linear regression of factors affecting data retention; experimental paradigm comparison
Key Findings: Across 766 sessions, younger infants were more likely to enter the scanner successfully; female infants and movie-based paradigms yielded more usable data than block/event designs. Social stimulus content outperformed abstract content. On average ~9 minutes of functional data were retained per session, and multi-experiment sessions increased overall yield. The study benchmarks practical parameters for labs establishing awake infant fMRI protocols.
DOI/URL: https://onlinelibrary.wiley.com/doi/10.1111/infa.70062

Paper 8
Title: Simultaneous EEG-fMRI investigation of sound sequence processing in human neonates
Authors: [bioRxiv consortium; full list at URL]
Journal/Venue: *bioRxiv* (preprint)
Date: January 10, 2026
Preprint? Yes (bioRxiv)
Age Range: Neonates (sleeping)
Primary Method: Simultaneous EEG + fMRI (auditory oddball paradigm)
Statistical Approach: Multimodal integration; ERP analysis; whole-brain fMRI GLM; network-level analysis of temporal, sensorimotor, frontal, thalamic, and hippocampal regions
Key Findings: Sleeping neonates show engagement of distributed cortical networks—spanning temporal, sensorimotor, and inferior frontal cortices—alongside subcortical regions (thalamus and hippocampus) during sound sequence deviance detection. The simultaneous EEG-fMRI approach revealed both temporal dynamics (via EEG) and spatial extent (via fMRI) of deviance processing, demonstrating that the neonatal brain already recruits a multi-hub auditory network in the first days of life.
DOI/URL: https://www.biorxiv.org/content/10.64898/2026.01.09.698718v1.full

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3.3 Developmental Trajectories

Paper 9
Title: Cascading periods of language-related brain plasticity across early childhood
Authors: Monica E. Ellwood-Lowe, Monami Nishio, Alexander J. Dufford, Michael Arcaro, Theodore D. Satterthwaite, Allyson P. Mackey
Journal/Venue: *bioRxiv* (preprint)
Date: March 27, 2026
Preprint? Yes (bioRxiv)
Age Range: 10 months–15 years (Baby Connectome Project: 10m–3.5y, n=222, 458 observations; HCP-Development: 5–15y, n=324)
Primary Method: fMRI (resting-state; Hurst exponent as index of cortical

plasticity)
Statistical Approach: Hurst exponent (long-range temporal correlations); growth curve modeling; cross-dataset validation; language skills regression
Key Findings: Cortical Hurst values increase in temporal and frontal language areas in early childhood, with posterior regions maturing before anterior regions; thalamic Hurst plateaus earlier. Counterintuitively, children with stronger language skills show slower increases in cortical Hurst in early childhood, consistent with protracted plasticity as an adaptive advantage. The cascade of sensitive periods across language-related cortical regions is confirmed across two independent large datasets spanning birth to adolescence.
DOI/URL: https://www.biorxiv.org/content/10.64898/2026.03.27.714739v1

Paper 10
Title: The emergence of the language system in the toddler brain
Authors: [bioRxiv preprint; full author list at URL]
Journal/Venue: *bioRxiv* (preprint)
Date: February 2026
Preprint? Yes (bioRxiv)
Age Range: 19–36 months (awake toddlers, N=29)
Primary Method: Awake task-based fMRI (language comprehension task)
Statistical Approach: Whole-brain GLM; ROI analysis of left frontal and temporal cortices; contrast: comprehensible vs. incomprehensible language
Key Findings: Regions in the left frontal and temporal cortex already show greater responses to comprehensible language in toddlers aged 19–36 months, demonstrating adult-like lateralization of language processing several years earlier than previously demonstrated with fMRI in awake children. The study confirms that foundational left-lateralized language networks are established during the toddler period.
DOI/URL: https://www.biorxiv.org/content/10.64898/2026.02.26.707550v1.full

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Paper 11
Title: Utilizing functional neuroimaging to study early language development
Authors: [Review article; Developmental Cognitive Neuroscience; multiple

authors]
Journal: *Developmental Cognitive Neuroscience* (ScienceDirect)
Date: 2025/2026 (online 2025; current issue 2026)
Preprint? No (peer-reviewed review)
Age Range: Infancy through toddlerhood (review scope: 0–36 months)
Primary Method: Review of fMRI, fNIRS, MEG, EEG approaches to language
Statistical Approach: Narrative review synthesizing ROI-based, whole-brain, and data-driven neuroimaging approaches
Key Findings: Spatially precise tools (fMRI, fNIRS) reveal topographic organization of language networks in infants, while temporally precise tools (EEG, MEG) track real-time language processing. The review identifies key future directions: longitudinal multi-modal designs, cross-method validation, and integration of environmental language input measures with neural outcomes. Understanding early language neurobiology can guide prediction of language delays and inform targeted early interventions.
DOI/URL: https://www.sciencedirect.com/science/article/pii/S1878929325001379

Paper 12
Title: Early-life neural correlates of behavioral inhibition and anxiety risk
Authors: [Review; Neuropsychopharmacology 2026; multiple authors]
Journal: *Neuropsychopharmacology*
Date: 2026
Preprint? No (peer-reviewed review)
Age Range: Infancy through early childhood (review scope)
Primary Method: EEG (resting-state; delta-beta coupling, frontal asymmetry)
Statistical Approach: Narrative review of rs-EEG measures; developmental cascade modeling; meta-analytic synthesis
Key Findings: Two rs-EEG measures—delta-beta coupling and frontal EEG asymmetry—show consistent associations with behavioral inhibition (BI) and anxiety risk from infancy. Infants with BI exhibit altered bottom-up attention and top-down control reflected in EEG during both rest and stimulus processing. The review argues for longitudinal, multimodal designs that integrate EEG, behavior, and neuroimaging to map risk trajectories from infancy through adolescence.
DOI/URL: https://www.nature.com/articles/s41386-025-02235-8

3.4 Oscillatory Activity & Neural Rhythms

(See also Papers 1, 2, 4, and 8 above, which primarily address oscillatory activity.)

Paper 13
Title: Infant sensory gating and a developmental cascade to autistic traits and anxiety
Authors: Rebecca F. Schwarzlose
Journal: *Neuropsychopharmacology* (Volume 51, Issue 1)
Date: January 2026
Preprint? No (peer-reviewed perspective/review)
Age Range: Neonatal through toddler period (review scope)
Primary Method: Review (EEG and fMRI sensory gating paradigms)
Statistical Approach: Narrative synthesis; developmental cascade framework; EEG P50 suppression and fMRI habituation measures
Key Findings: Impaired sensory gating during a sensitive neonatal period promotes a phenotype of sensory over-responsivity, autistic traits, anxiety, and broader psychiatric challenges. EEG and fMRI provide complementary temporal and spatial resolution for measuring sensory gating non-invasively in infants. The paper proposes a mechanistic cascade model linking early-life sensory filtering failures to later neurodevelopmental and psychiatric outcomes, with implications for screening.
DOI/URL: https://www.nature.com/articles/s41386-025-02253-6

3.5 Autism Spectrum Disorder

Paper 14
Title: Functional connectivity in infants' visual cortex and its links to motion processing and autism
Authors: [Multiple; Scientific Reports 2026]

Journal: *Scientific Reports* (Nature Publishing Group)
Date: March 2026
Preprint? No (peer-reviewed)
Age Range: Infants (first year of life; at-risk for autism cohort)
Primary Method: EEG (dbWPLI functional connectivity; visual cortical gamma)
Statistical Approach: Debiased weighted phase lag index (dbWPLI); regression of connectivity on later autism symptoms and global motion laterality; longitudinal design
Key Findings: Gamma-band EEG connectivity between midline and far-lateral visual cortex during social scene viewing predicted later autism symptoms and global motion visual cortical laterality. EEG visual cortical connectivity is thus linked to both later autism outcome and atypical motion processing, suggesting a shared neural mechanism. The dbWPLI method is highlighted as particularly suited to infant EEG due to its resistance to volume conduction artifacts.
DOI/URL: https://www.nature.com/articles/s41598-026-42048-3

Paper 15
Title: The association between infant EEG aperiodic exponent and the trajectory of restricted and repetitive behaviors for toddlers with and without autism
Authors: [Multiple; Journal of Neurodevelopmental Disorders]
Journal: *Journal of Neurodevelopmental Disorders* (Springer Nature)
Date: 2025 (published late 2025/early 2026)
Preprint? No (peer-reviewed)
Age Range: 12–36 months (EEG at 12–14 months; RRBs followed to 36 months)
Primary Method: EEG (resting-state; aperiodic exponent as E/I proxy)
Statistical Approach: FOOOF aperiodic decomposition; longitudinal growth curve modeling of RRBs (Repetitive Behavior Scale-Revised); group comparisons (autism vs. non-autism elevated-likelihood)
Key Findings: Lower aperiodic exponent at 12–14 months was associated with elevated and increasing restricted and repetitive behaviors from 12 to 36 months; this association was specific to RRBs and not to general cognitive or verbal development. The E/I specificity to RRBs positions the aperiodic exponent as a domain-specific autism biomarker, distinct from general

developmental lag.
DOI/URL: https://link.springer.com/article/10.1186/s11689-025-09651-3

Paper 16
Title: Changes in Early Aperiodic EEG Activity Are Linked to Autism Diagnosis and Language Development in Infants With Family History of Autism
Authors: Wilkinson et al.
Journal: *Autism Research* (Wiley)
Date: 2025
Preprint? No (peer-reviewed)
Age Range: 3–12 months (longitudinal; language outcomes at 18 months)
Primary Method: EEG (resting-state; aperiodic offset and slope)
Statistical Approach: Mixed-effects growth modeling of aperiodic parameters from 3 to 12 months; logistic regression for autism diagnosis; language outcome regression (18-month measures)
Key Findings: Infants with elevated familial likelihood (EFL) for autism show significantly lower aperiodic activity from 6.7–55 Hz at 3 months. Rate of change in aperiodic activity from 3 to 12 months was significantly greater in infants later diagnosed with autism versus EFL non-diagnosed peers. Greater increases in aperiodic offset and slope were additionally associated with worse language outcomes at 18 months.
DOI/URL: https://onlinelibrary.wiley.com/doi/10.1002/aur.70063

Paper 17
Title: Machine learning and deep learning applied to EEG and fNIRS for early autism spectrum disorder diagnosis: a systematic review
Authors: [Multiple; Frontiers in Psychiatry editorial team and contributors]
Journal: *Frontiers in Psychiatry*
Date: 2026
Preprint? No (peer-reviewed systematic review)
Age Range: Infants through children (studies from 2019–2024 reviewed)
Primary Method: Systematic review of ML/DL applied to EEG and fNIRS
Statistical Approach: PRISMA-compliant review; accuracy/sensitivity/specificity meta-summary across 27 studies

Key Findings: Mean classification accuracy across 27 included studies was 93.39% (SD=6.50%), demonstrating high discriminative performance of EEG- and fNIRS-based ML/DL for ASD detection. EEG (millisecond-level temporal resolution) and fNIRS (hemodynamic tracking) are shown to be complementary modalities. The review identifies a critical need for larger pediatric validation datasets and age-stratified performance metrics.
DOI/URL: https://www.frontiersin.org/journals/psychiatry/articles/10.3389/fpsyt.2026.1668914/full

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3.6 Down Syndrome

Paper 18
Title: Single-cell atlas of the developing Down syndrome brain cortex
Authors: Lattke M., et al. (Duke-NUS Medical School; Imperial College London; international consortium)
Journal: *Nature Medicine*
Date: January 16, 2026
Preprint? No (peer-reviewed; earlier bioRxiv preprint: December 29, 2025)
Age Range: 10–20 weeks postconception (fetal cortex)
Primary Method: Single-cell multiome: simultaneous single-nucleus RNA-seq + ATAC-seq (~250,000 cells); human neural progenitor antisense oligonucleotide validation; humanized mouse model
Statistical Approach: Single-cell clustering; differential cell-type abundance analysis; transcriptional network analysis; chromatin accessibility profiling; antisense oligonucleotide functional rescue assay
Key Findings: Subtype-specific reductions in RORB- and FOXP1-expressing excitatory neurons and widespread disruption of neurodevelopmental transcriptional programs were identified in DS cortex. Three chromosome 21 transcription factors—BACH1, PKNOX1, and GABPA—emerged as dosage-sensitive regulatory hubs linked to intellectual disability gene networks. Antisense oligonucleotide normalization of these factors partially rescued target gene expression in human neural progenitors in vitro.
DOI/URL: https://www.nature.com/articles/s41591-026-04211-1 | PubMed: https://pubmed.ncbi.nlm.nih.gov/41545595/

3.7 Multivariate Analyses

(Papers 5, 6, 9, and 17 above represent the primary multivariate/machine learning contributions. Additional note below.)

Cortical markers of E/I balance and sensory responsivity *(Translational Psychiatry, December 2025)*: A prospective longitudinal cohort of 151 infants with/without familial autism or ADHD history used EEG at 5, 10, and 14 months to track aperiodic-based E/I markers. Distinct trajectory patterns were found: autism family history predicted early increases in sensory hyper-responsivity; ADHD family history predicted increasing hypo-responsivity. Multivariate mixed-effects trajectory modeling separated clinical-group-specific developmental paths from general neurodivergence markers. DOI: https://www.nature.com/articles/s41398-025-03791-9

4. Methods Overview

Method	# Papers	% of Total
EEG (resting-state / oscillatory)	9	50%
fMRI (resting-state)	4	22%
fMRI (task-based / awake)	3	17%
Simultaneous EEG+fMRI	1	6%
Single-cell genomics	1	6%
Systematic review / ML	1	6%
Multimodal / review	3	17%

(Note: several papers use multiple methods; percentages reflect primary method classification.)

5. Statistical Approaches Trending This Month

1. **FOOOF / specparam aperiodic decomposition**: The dominant analytical innovation this month. Multiple independent groups are now decomposing EEG power spectra into aperiodic (1/f slope and offset) and periodic (oscillatory peak) components. Aperiodic parameters are being used as proxies for excitation/inhibition (E/I) balance across autism risk, nutrition, maternal mental health, and normative development studies.

2. **Connectome-based predictive modeling (CPM)**: Applied to neonatal resting-state fMRI to prospectively predict 18-month socioemotional outcomes. Stability-driven, ROI-constrained CPM is becoming a standard for prognostic functional connectivity studies in early life.

3. **Hurst exponent analysis of fMRI timeseries**: Used to quantify in vivo cortical plasticity / long-range temporal correlations. The application to large longitudinal pediatric datasets (Baby Connectome Project, HCP-D) to map sensitive periods is novel and growing.

4. **Nonlinear functional connectivity analysis**: Emerging as a complement to standard linear ICA and seed-based approaches in infant fMRI, with the first whole-brain application to infant developmental trajectories reported this month.

5. **Longitudinal growth curve / mixed-effects modeling**: Standard for developmental EEG trajectory studies; increasingly applied to model both linear and nonlinear age-related changes in power, aperiodic parameters, and connectivity.

6. **Deep learning / machine learning applied to infant EEG and fNIRS**: Systematic review confirms mean accuracy >93% for ASD classification; field is moving toward age-stratified, explainable AI frameworks.

7. **debiased Weighted Phase Lag Index (dbWPLI)**: Recommended for infant EEG connectivity to minimize spurious short-range connectivity due to small head size and volume conduction.

6. Notable Authors & Labs This Month

Researcher	Institution	Specialty	Paper
Allyson P. Mackey	University of Pennsylvania	Plasticity & sensitive periods	Cascading language plasticity (bioRxiv March 2026)
Monica E. Ellwood-Lowe	UC Berkeley	Developmental fMRI, language	Cascading language plasticity (bioRxiv March 2026)
Theodore D. Satterthwaite	University of Pennsylvania	Developmental neuroimaging	Cascading language plasticity (bioRxiv March 2026)
Rebecca F. Schwarzlose	Washington University in St. Louis	Sensory processing, autism	Sensory gating and autism cascade (Neuropsychopharmacology 2026)
Wilkinson et al.	UK Autism Research collaborators	EEG, autism biomarkers	Aperiodic EEG linked to autism & language (Autism Research 2025)
Lattke M. et al.	Duke-NUS / Imperial College London	Developmental molecular neuroscience	Single-cell atlas Down syndrome cortex (Nature Medicine 2026)
Gilbreath M. et al.	Nutrition & neuroscience	Infant nutrition × brain	Nutrition and aperiodic exponent (Psychophysiology 2026)
Developing Human Connectome Project Consortium	King's College London / international	Neonatal MRI	Neonatal fMRI predicts 18-month outcomes (bioRxiv May 2026)

Researcher	Institution	Specialty	Paper
Behm et al.	Yale + MIT	Awake infant fMRI methodology	>750 awake fMRI sessions (Infancy 2026)
Beaugrand M., Jaramillo V., Mühlematter C.	Swiss (CHUV/ UNIL)	Infant sleep EEG	Sleep neurophysiology 3–6 months (npj 2026)

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7. Preprints to Watch

The following papers on bioRxiv / PsyArXiv are not yet peer-reviewed and represent the leading edge of the field:

1. **Neonatal Resting-State Functional Connectivity Predicts Socioemotional and Behavioral Outcomes at 18 Months** (bioRxiv, May 05, 2026) — High-impact potential; connectome-based predictive modeling of socioemotional temperament from neonatal fMRI. Watch for peer-reviewed journal submission in coming weeks.
URL: https://www.biorxiv.org/content/10.64898/2026.04.21.719787v2

2. **Explicitly nonlinear fMRI networks reveal hidden trajectories of infant brain development** (bioRxiv, April 08, 2026) — Methodological landmark; nonlinear connectivity analysis in infant fMRI. Likely to generate significant follow-up work and methodological adoption.
URL: https://www.biorxiv.org/content/10.64898/2026.04.07.716703v1

3. **Cascading periods of language-related brain plasticity across early childhood** (bioRxiv, March 27, 2026) — High-quality dataset (Baby Connectome Project + HCP-D); Hurst exponent application to map language sensitive periods across development. Major implications for educational and intervention timing.
URL: https://www.biorxiv.org/content/10.64898/2026.03.27.714739v1

4. **The emergence of the language system in the toddler brain** (bioRxiv, February 2026) — First awake toddler fMRI demonstration of left-lateralized language responses; important for establishing timeline of language network formation.
URL: https://www.biorxiv.org/content/10.64898/2026.02.26.707550v1.full

5. **Simultaneous EEG-fMRI investigation of sound sequence processing in human neonates** (bioRxiv, January 2026) — Technically demanding simultaneous EEG-fMRI in sleeping neonates; results challenge assumptions about neonatal cortical auditory processing extent.
URL: https://www.biorxiv.org/content/10.64898/2026.01.09.698718v1.full

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8. Complete Paper List

Full Annotated Bibliography

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[1] Beaugrand M., Jaramillo V., Mühlematter C., et al. (2026). *Tracing infant sleep neurophysiology longitudinally from 3 to 6 months: EEG insights into brain development.* npj Biological Timing and Sleep. Published March 2026.
- Age: 3–6 months | Method: EEG (NREM sleep) | Approach: longitudinal EEG power, topographic analysis
- Key: SWA increases occipitally; theta globally; sigma disperses frontally — consistent with thalamocortical maturation
- URL: https://www.nature.com/articles/s44323-026-00071-7 | Preprint: No

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[2] Gilbreath M., et al. (2026). *Effects of Early Infant Nutrition on Aperiodic Exponent From 2 to 12 Months Using EEG Analysis.* Psychophysiology. Published February 2026. DOI: 10.1111/psyp.70247
- Age: 2–12 months | Method: High-density EEG | Approach: FOOOF/specparam, source reconstruction, mixed models
- Key: Aperiodic exponent decreases with age; breastfed infants show larger

exponent at 3 months only
- URL: https://onlinelibrary.wiley.com/doi/10.1111/psyp.70247 | Preprint: No

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[3] [Author list TBC]. (2026). *EEG trajectories across the first two years of life are associated with maternal depression, anxiety, and perceived stress.* Journal of Affective Disorders.
- Age: 2–24 months (N=211) | Method: EEG (β and γ bands) | Approach: Longitudinal mixed models, maternal mental health scales
- Key: Higher maternal distress → lower infant beta/gamma power trajectories across 2 years
- URL: https://www.sciencedirect.com/science/article/abs/pii/S0165032725020294 | Preprint: No

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[4] [Author list TBC]. (2026). *Resting-state periodic and aperiodic brain oscillations from birth to preschool years: Aperiodic maturity predicts developmental course.* Developmental Cognitive Neuroscience.
- Age: 2–68 months | Method: EEG (source-localized, 7 regions) | Approach: Specparam, nonlinear growth modeling, adaptive behavior regression
- Key: Aperiodic maturity at earlier ages predicts better adaptive outcomes; region-specific trajectories established
- URL: https://www.sciencedirect.com/science/article/pii/S1878929326000447 | Preprint: No

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[5] Developing Human Connectome Project Consortium. (2026). *Neonatal Resting-State Functional Connectivity Predicts Socioemotional and Behavioral Outcomes at 18 Months.* bioRxiv. Posted May 05, 2026.
- Age: Neonates → 18 months (N=397: 277 term, 120 preterm) | Method: Resting-state fMRI | Approach: Stability-driven ROI-constrained CPM; CBCL; ECBQ
- Key: Neonatal connectivity predicts externalizing, surgency, effortful control at 18 months; term vs. preterm architectures differ
- URL: https://www.biorxiv.org/content/10.64898/2026.04.21.719787v2 | Preprint: Yes (bioRxiv)

[6] [Author list TBC]. (2026). *Explicitly nonlinear fMRI networks reveal hidden trajectories of infant brain development.* bioRxiv. Posted April 08, 2026.
- Age: Typically developing infants | Method: Resting-state fMRI | Approach: Data-driven nonlinear connectivity decomposition; developmental regression
- Key: Nonlinear networks reveal DMN, executive, language production networks invisible to linear methods
- URL: https://www.biorxiv.org/content/10.64898/2026.04.07.716703v1 | Preprint: Yes (bioRxiv)

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[7] Behm T., et al. (2026). *Functional Magnetic Resonance Imaging in Awake Infants: Insights from More Than 750 Scanning Sessions.* Infancy. Published 2026. DOI: 10.1111/infa.70062
- Age: 1–36 months (N=766 sessions; Yale + MIT) | Method: Awake task-based fMRI | Approach: Multi-lab retrospective logistic/linear regression
- Key: Younger age, female sex, movies, social stimuli → better data yield; ~9 min average usable data
- URL: https://onlinelibrary.wiley.com/doi/10.1111/infa.70062 | Preprint: No

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[8] [Author list TBC]. (2026). *Simultaneous EEG-fMRI investigation of sound sequence processing in human neonates.* bioRxiv. Posted January 10, 2026.
- Age: Neonates (sleeping) | Method: Simultaneous EEG + fMRI | Approach: ERP analysis + whole-brain fMRI GLM; auditory oddball paradigm
- Key: Multi-hub auditory network (temporal, sensorimotor, inferior frontal, thalamus, hippocampus) active in sleeping neonates for deviance detection
- URL: https://www.biorxiv.org/content/10.64898/2026.01.09.698718v1.full | Preprint: Yes (bioRxiv)

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[9] Ellwood-Lowe M.E., Nishio M., Dufford A.J., Arcaro M., Satterthwaite T.D., Mackey A.P. (2026). *Cascading periods of language-related brain plasticity across early childhood.* bioRxiv. Posted March 27, 2026.
- Age: 10 months–15 years (BCP n=222; HCP-D n=324) | Method: Resting-

state fMRI (Hurst exponent) | Approach: Growth curve modeling, cross-dataset validation, language skills regression
- Key: Posterior-to-anterior cascade of language sensitive periods; higher skills → slower Hurst increases (protracted plasticity)
- URL: https://www.biorxiv.org/content/10.64898/2026.03.27.714739v1 | Preprint: Yes (bioRxiv)

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[10] [Author list TBC]. (2026). *The emergence of the language system in the toddler brain.* bioRxiv. Posted February 2026.
- Age: 19–36 months (N=29 awake toddlers) | Method: Awake task-based fMRI | Approach: Whole-brain GLM; language > baseline contrast; left lateralization analysis
- Key: Left frontal and temporal cortex already prefer comprehensible language in toddlers 19–36 months
- URL: https://www.biorxiv.org/content/10.64898/2026.02.26.707550v1.full | Preprint: Yes (bioRxiv)

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[11] [Multiple authors]. (2025/2026). *Utilizing functional neuroimaging to study early language development.* Developmental Cognitive Neuroscience.
- Age: 0–36 months (review scope) | Method: Review (fMRI, fNIRS, EEG, MEG) | Approach: Narrative synthesis
- Key: Comprehensive review of neuroimaging approaches to early language; identifies multimodal longitudinal designs as priority
- URL: https://www.sciencedirect.com/science/article/pii/S1878929325001379 | Preprint: No

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[12] [Multiple authors]. (2026). *Early-life neural correlates of behavioral inhibition and anxiety risk.* Neuropsychopharmacology.
- Age: Infancy through early childhood (review scope) | Method: EEG review (delta-beta coupling; frontal asymmetry) | Approach: Narrative meta-analytic review
- Key: rs-EEG measures link to behavioral inhibition from infancy; delta-beta

coupling and frontal asymmetry as key risk indices
- URL: https://www.nature.com/articles/s41386-025-02235-8 | Preprint: No

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[13] Schwarzlose R.F. (2026). *Infant sensory gating and a developmental cascade to autistic traits and anxiety.* Neuropsychopharmacology, 51(1), 86–94.
- Age: Neonatal–toddler (review scope) | Method: EEG/fMRI sensory gating review | Approach: Mechanistic cascade model; developmental perspective
- Key: Neonatal sensory gating failure → sensory over-responsivity → autism traits + anxiety; cascade model proposed
- URL: https://www.nature.com/articles/s41386-025-02253-6 | Preprint: No

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[14] [Multiple authors]. (2026). *Functional connectivity in infants' visual cortex and its links to motion processing and autism.* Scientific Reports.
- Age: First year of life (autism-risk cohort) | Method: EEG (dbWPLI gamma connectivity) | Approach: dbWPLI; autism symptom regression; global motion laterality analysis
- Key: Visual cortex gamma connectivity predicts later autism symptoms and motion processing lateralization
- URL: https://www.nature.com/articles/s41598-026-42048-3 | Preprint: No

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[15] [Multiple authors]. (2025). *The association between infant EEG aperiodic exponent and the trajectory of restricted and repetitive behaviors for toddlers with and without autism.* Journal of Neurodevelopmental Disorders. DOI: 10.1186/s11689-025-09651-3
- Age: 12–36 months (EEG at 12–14 months; RRBs at 12, 24, 36 months) | Method: EEG (aperiodic exponent) | Approach: FOOOF; longitudinal RRBs growth modeling; specificity analysis vs. cognition
- Key: Lower aperiodic exponent at 12 months → increasing RRBs through toddlerhood; independent of cognitive trajectory
- URL: https://link.springer.com/article/10.1186/s11689-025-09651-3 | Preprint: No

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[16] Wilkinson et al. (2025). *Changes in Early Aperiodic EEG Activity Are Linked to Autism Diagnosis and Language Development in Infants With Family History of Autism.* Autism Research. DOI: 10.1002/aur.70063
- Age: 3–12 months longitudinal; language outcomes at 18 months | Method: EEG (aperiodic offset and slope) | Approach: Mixed-effects longitudinal; autism diagnosis logistic regression; 18-month language regression
- Key: EFL infants diagnosed with autism show greater 3→12 month aperiodic changes; changes also predict worse language at 18 months
- URL: https://onlinelibrary.wiley.com/doi/10.1002/aur.70063 | Preprint: No

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[17] [Multiple authors; Frontiers]. (2026). *Machine learning and deep learning applied to EEG and fNIRS for early autism spectrum disorder diagnosis: a systematic review.* Frontiers in Psychiatry. DOI: 10.3389/fpsyt.2026.1668914
- Age: Infants–children (27 studies, 2019–2024) | Method: Systematic review of ML/DL on EEG + fNIRS | Approach: PRISMA; accuracy meta-summary; subgroup analyses by modality
- Key: Mean accuracy 93.39%; EEG and fNIRS are complementary; need for larger age-stratified validation datasets
- URL: https://www.frontiersin.org/journals/psychiatry/articles/10.3389/fpsyt.2026.1668914/full | Preprint: No

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[18] Lattke M., et al. (2026). *Single-cell atlas of the developing Down syndrome brain cortex.* Nature Medicine. Published January 16, 2026. PubMed: 41545595
- Age: 10–20 weeks postconception (fetal, N=15 DS, 15 control) | Method: Single-cell multiome (scRNA-seq + ATAC-seq, ~250K cells) | Approach: Single-cell clustering; differential abundance; regulatory network analysis; ASO rescue experiments
- Key: RORB/FOXP1 excitatory neuron reduction; BACH1/PKNOX1/GABPA as chr21 dosage-sensitive TF hubs; partial ASO rescue demonstrated
- URL: https://www.nature.com/articles/s41591-026-04211-1 | PubMed: https://pubmed.ncbi.nlm.nih.gov/41545595/ | Preprint: No

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